SECTION 9

OBSERVE

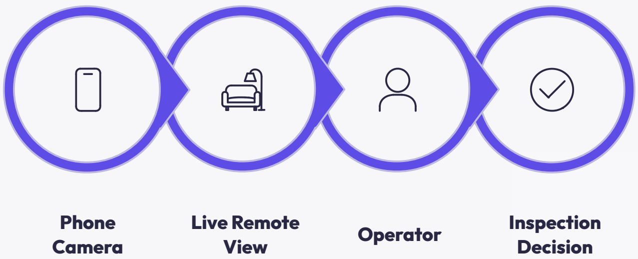
A REMOTE EYE INTO THE ENVIRONMENT.

The CyberRover X4 is equipped with a phone camera that provides live visual observation of the rover's surroundings. The operator views the camera feed in real time, enabling remote inspection of environments that are difficult or dangerous to access directly.

Current Visual Capabilities

- **Live Remote Camera Feed**
- **Visual Confirmation of Obstacles and Terrain**
- **Observation of Hazardous Locations**
- **Support for Navigation and Inspection Decisions**

Vision Pipeline



Future Development: A dedicated analog drone camera can replace or supplement the phone-camera arrangement in a future version, providing a more integrated onboard vision system with improved stability and field-of-view control.

SECTION 10

THE OPERATOR

CONTROL WITHOUT DIRECT EXPOSURE.

The CyberRover X4 is controlled via a custom ESP32-based remote controller using ESP-NOW wireless protocol. Commands flow to an Arduino Uno on the rover, keeping the operator safe while extending access into hazardous environments.

ESP-NOW Wireless Link

Joystick Input

OLED Cyber OS Display

Real-Time Telemetry

Safety-Oriented Behavior

The remote controller operates silently with buzzer disabled by design, and features dedicated **Back**, **Enter**, and **Save** button controls integrated into the Cyber OS interface.

SECTION 11

DEPLOYMENT

WHERE CYBERROVER X4 CAN GO FIRST.

Industrial Inspection

Gas-leak detection, hazardous atmospheres, machinery areas, restricted zones.

Confined Spaces

Pipes, ducts, crawl spaces, tunnels, narrow passages.

Mining & Hazardous Environments

Remote inspection where gases, heat, or difficult terrain create additional risk.

Fire-Affected Areas

Remote visual and environmental assessment in or near fire-affected locations.

Earthquake & Disaster Response

Inspect damaged structures and debris zones before human responders enter.

Remote Reconnaissance

Observation of distant, uncertain, or difficult-to-access locations.

Note: Army and defence surveillance applications are potential use cases. The CyberRover X4 is not currently deployed in military operations or certified for battlefield use.

SECTION 12

LIMITATIONS

EVERY ENGINEERING SYSTEM HAS BOUNDARIES.

The CyberRover X4 is a prototype — understanding its current limitations is essential for safe deployment and future development.

Environmental Sensing

MQ sensors provide indication only — not certified gas measurements.

Communication

Wireless only — subject to range limits, interference, signal loss.

Temperature

DHT11 & BMP280 not rated for extreme heat environments.

Mobility & Power

Terrain, traction, chassis, and battery constrain operational reach.

Ultrasonic Sensing

Three sensors with blind spots, limited range, no 3D mapping.

Autonomy & Intelligence

Fully operator-controlled — no AI, computer vision, or autonomy.

Visual Inspection

Phone camera — limited FOV, lighting sensitivity, mounting constraints.

Safety & Certification

Not certified as explosion-proof, intrinsically safe, or military-grade.

SECTION 13

ENGINEERING STATUS

ENGINEERED IN LAYERS. BUILT TO EVOLVE.

Built & Demonstrated

- 4WD mobility
- BTS7960 motor drivers
- ESP32-S3 N8R16
- Arduino Uno motor/ultrasonic controller
- MQ-4, MQ-7, MQ-135 sensors
- DHT11 & BMP280 (Pressure/Altitude)
- 3× ultrasonic sensors
- Phone-camera visual
- OLED Cyber OS
- ESP32-CAM (Wi-Fi Telemetry Hub)
- 16x2 I2C LCD local display
- Acoustic Tactical Siren (10 sound profiles)

Integration & Refinement

- System reliability
- Sensor integration
- Telemetry robustness
- Field testing
- Mechanical refinement
- Environmental testing
- Power management
- Wireless range
- Sensor calibration
- Durability validation

Future Development

- Industrial high-temp sensing
- Dedicated analog camera
- Integrated vision system
- Computer vision
- Voice interface
- AI command interpretation
- Autonomous navigation
- Advanced obstacle avoidance
- Extended sensor suite
- Modular payloads

1

Current Prototype

2

Field Validation

3

Hardened Hardware

4

Advanced Vision

5

AI / Autonomy

SECTION 14

MOVE. SENSE. INSPECT.

CYBERROVER X4.2

EXTENDING HUMAN AWARENESS INTO PLACES THAT ARE DIFFICULT, DANGEROUS OR IMPOSSIBLE TO INSPECT DIRECTLY.
